

ELORIE BERNARD-LACROIX

eb.lacroix04@gmail.com · Toronto, ON

Website: elorie-bernard-lacroix.github.io · LinkedIn: [elorie-bernard-lacroix](#) · GitHub: [elorie-bernard-lacroix](#)

EDUCATION

Bachelor of Applied Science (BASc) in Engineering Science + PEY

Major in Machine Intelligence, Minor in Environmental Engineering

University of Toronto

Sept 2022 – May 2027

Toronto, ON

- **Relevant Courses:** Artificial Intelligence, Matrix Algebra & Optimization, Thermodynamics, Environmental Impact & Risk Assessment, Economic Analysis & Decision Making, Fundamentals of Electric Circuits
- **Community:** Iron Dragons, Concrete Canoe (software), Robotics for Space Exploration (simulation software)
- **Awards:** Procter & Gamble scholarship (\$8000), Dean's Merit Award (\$5000), Faculty of Applied Science and Engineering Admission Scholarship (\$5000)

PROFESSIONAL EXPERIENCE

Software Engineering Intern

Procter & Gamble

May 2025 – Present

North York, ON

- Coordinated between operation teams, product teams, and consulting teams to develop and maintain **8 production-grade** data validation pipelines between **3 applications** (Salesforce, Azure, SharePoint) using **Databricks** (Python, SQL), **Azure Data Factory**, and **Power BI**. Validated **800K records daily**, while keeping pipelines **modular** for all project phases and releases.
- Developed and operated **7 web-scraping automations** in Python (Selenium, Pandas) with **multi-processing** and **automated file tracking**, archiving **20 GB of data** and saving **\$100K** in vendor costs.
- Collaborated with 2 other interns to develop an automation in Python that consolidates **10 dynamic sources of data** and provides automatic multi-page summaries of data alignment between two main systems, saving over **3000 hours a year** and improving data accuracy for more than **70 users**.

Test Automation and Vehicle Instrumentation Intern

Bombardier Recreational Products (French-speaking Environment)

May 2024 – Aug 2024

Valcourt, QC

- Developed a **Python package** that automates a **battery test bench** in accordance with stakeholder needs and **IEC standards** (along with technical documentation and flow charts), allowing for more precise testing and significant time savings.
- Read and created new **electrical schematics**, **mechanical drawings**, bills of materials, and **budgets** to troubleshoot and maintain **3 towing dynamometers** (trailers equipped with an Eddy current brake to apply a load to vehicles). Created a Google Sheets tool with **Apps Script** to facilitate the maintenance and operation of equipment.
- Led 2 main projects (above), **defining their scope** and **requirements**, along with their respective tasks and deliverables.
- Led a training session with over **30 stakeholders**, demonstrating the operation of newly developed tools and ensuring **seamless integration** into existing workflows.

Information Technology Auditing Intern

Treasury Board Secretariat of the Ontario Public Service

May 2023 – Aug 2023

Toronto, ON

- Designed presentations in **Power BI** for a team of over **10 people**.
- Compiled **50 extracts** of research on talent management in interactive Excel sheets with different views, including pivot tables and filtering options to assist the team in an upcoming audit.
- Used standards provided by **ISO, COBIT, NIST, and GO-ITS** to supplement research and guide audit recommendations.
- Recorded meeting notes for **client interviews**, managed documents, proofread the audit report, and performed research to assist in a 3-month audit of **End-User IT Asset Lifecycle Management**.

Information Technology Technician

Canada's Wonderland

Apr 2022 – Aug 2022

Vaughan, ON

- Collaborated with 3 other technicians to perform upgrades to **Simphony 2.0** and deploy over **300 POS systems** in the park, including their respective **peripherals** (scanners, printers, cameras) and required **peripheral drivers**, to ensure a fully operational cashless park.
- Troubleshoot **networking**, **hardware**, and **software** issues for employees in the office.
- Configured **Cisco network phones** and **Cisco Finesse** (queues shared among phones) to allow for more effective online customer service support. Carefully **documented** this process for future technicians.

PROJECTS

Canoe Generator (UofT Concrete Canoe Team) Python, Multi-Objective Optimization [More Information](#)

Worked with other Hull & Structural leads in the UofT Concrete Canoe design team to build a Python program that searches a given space of canoe parameters and optimizes for target values of drag, stability, and more.

SmartStudy, AI Habit Optimizer Python, Machine Learning, GenAI, Bayesian Optimization [More Information](#)

Collaborated with two other students to build SmartStudy, an AI-driven habit optimizer that uses TabPFN and Bayesian optimization to predict and maximize student GPAs. Used k-Nearest Neighbors to surface peer-based behavioral insights and integrated GPT-4 to deliver natural language academic coaching. Also experimented with XGBoost, 1D CNN, and mixed models.

Automated Water Sanitation System Python, Mechatronics, CAD [More information](#)

Collaborated with 4 other students to prototype a system that could heat water to a specified temperature for a given time and then automatically dispense it to a tank of potable water. This project involved the entire design process from problem definition, to rapid prototyping (CAD, 3D printing) and testing.

Seam-Carving Program C, Dynamic Programming [More Information](#)

Designed a program that iteratively removes parts of an image that are the least important to narrow the image without distortion.

Predicting Failure Load MATLAB [More Information](#)

Built a MATLAB program that simulates a given bridge design and calculates its factor of safety against various forms of failure. Result: 3rd place.

SKILLS

- **Languages:** Python, C, Java, MATLAB, SQL, R, HTML/CSS, JavaScript
- **Data & AI:** PyTorch, TensorFlow, Scikit-learn, Deep Learning (Transformers, CNN, RNN, GNN), Reinforcement Learning (Q-Learning), Unsupervised Learning (GMM, kNN), Bayesian Optimization
- **Systems & Robotics:** ROS, Databricks, Azure Cloud, Linux (Ubuntu), Control Systems (Arduino, Raspberry Pi), Computer Vision, CAD (OnShape, Fusion 360), Unity
- **Tools & Workflow:** Git/GitHub, Docker, VS Code, Virtual Environments (Conda/Pip), Power BI, Pandas, NumPy, Matplotlib, Selenium, LaTeX